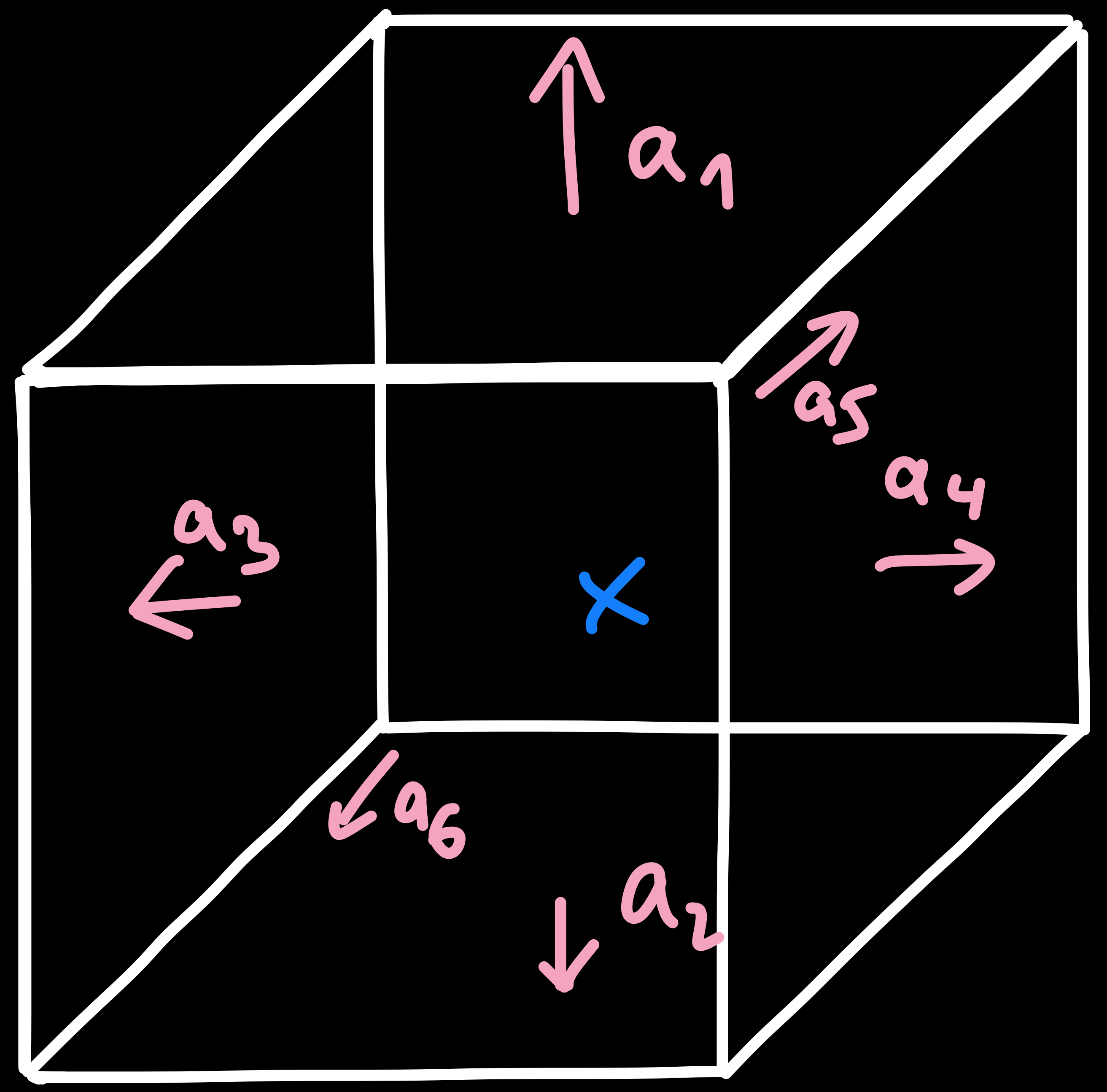
On the complexity of the simplex method

Eleon Bach

maximize $c^T x$

subject to $Ax \le b$

$$A = \begin{pmatrix} - & a_1 & - \\ & \vdots & \\ - & a_n & - \end{pmatrix}$$



38. Mathematical Tables Project computers with adding machines

David A. Grier, "When computers were human."

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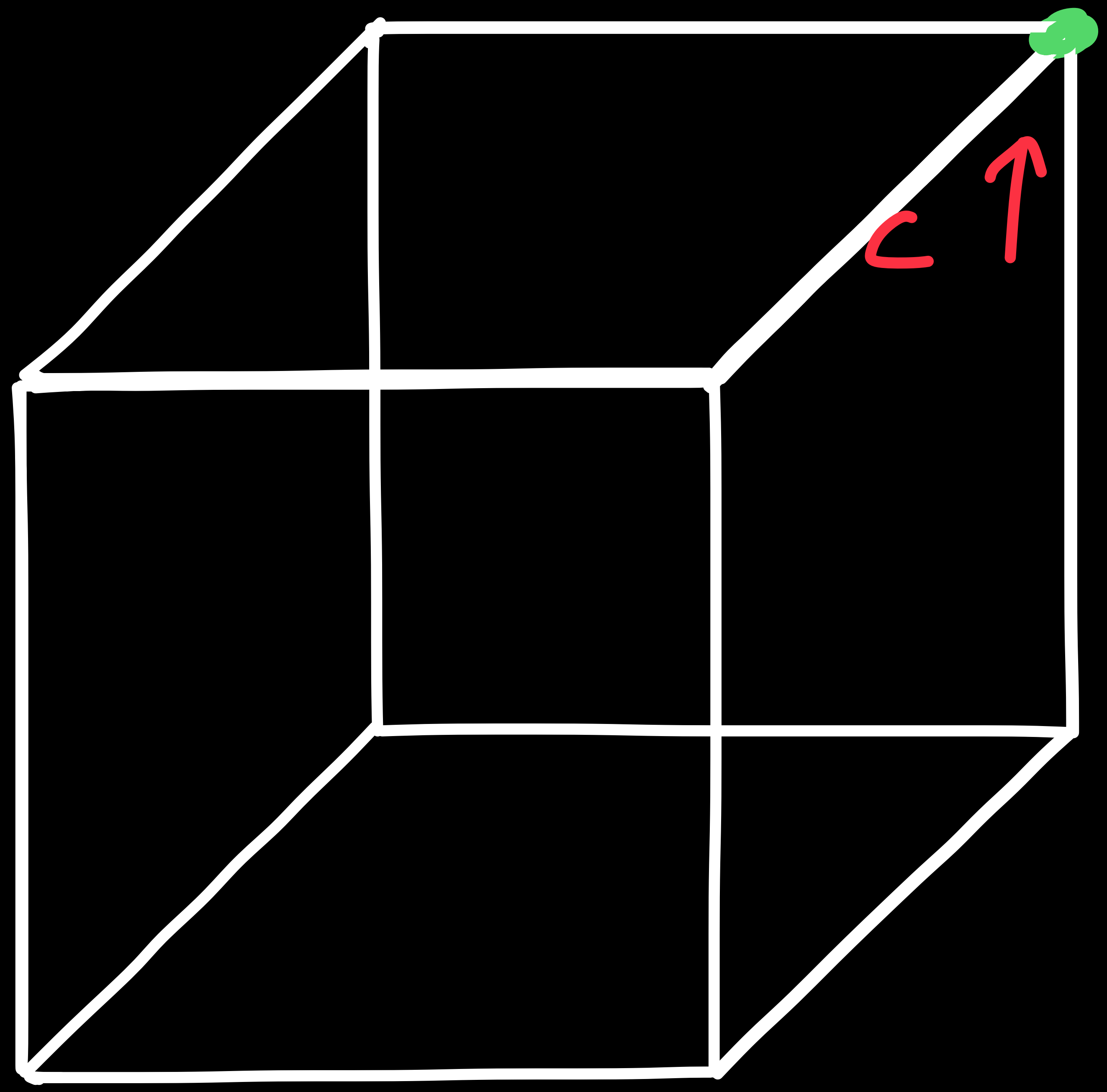
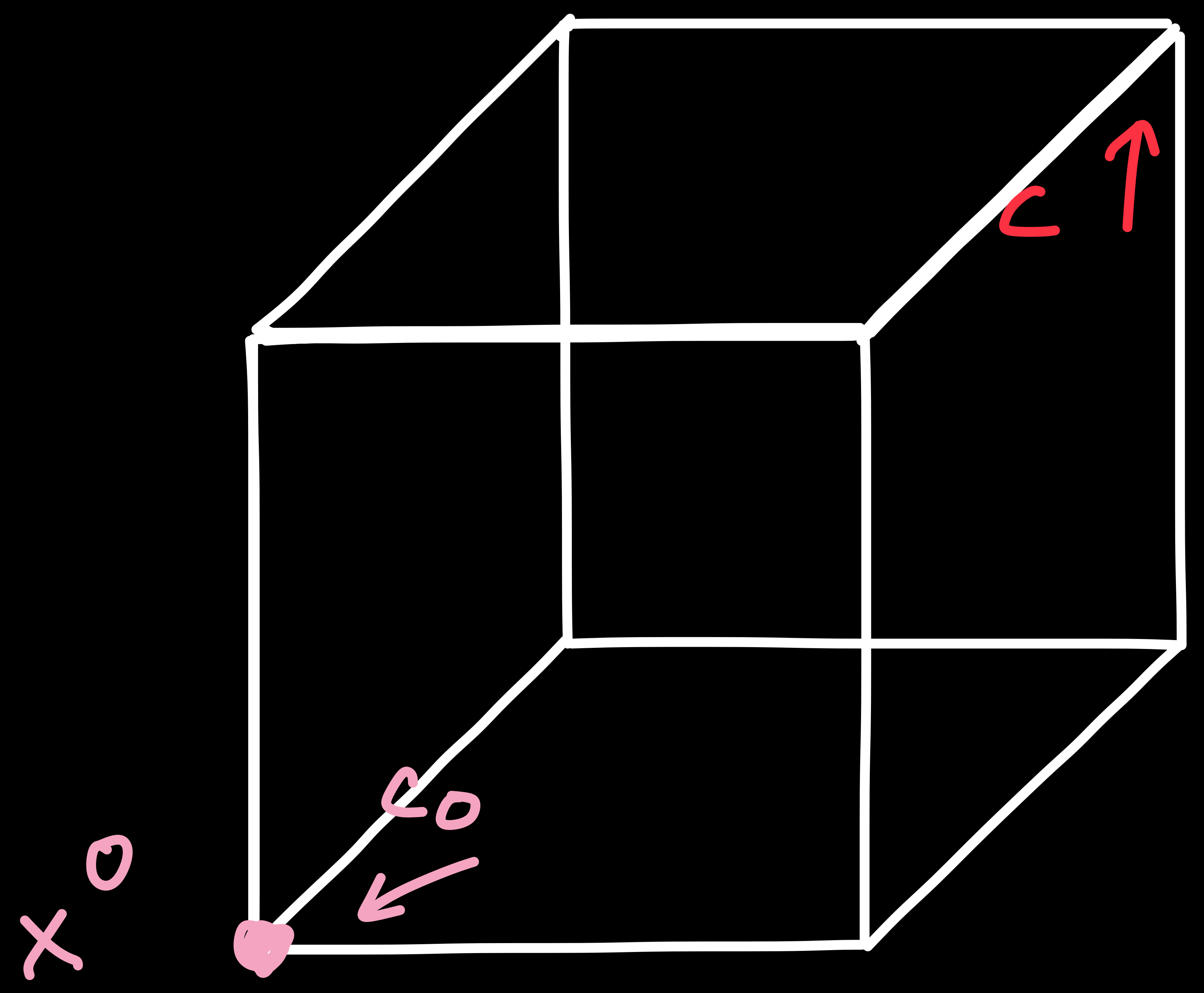
$$A = \begin{pmatrix} - & a_1 & - \\ & \vdots & \\ - & a_n & - \end{pmatrix}$$

$$\text{opt} = \max_{\{x : Ax \leq b\}} c^T x$$

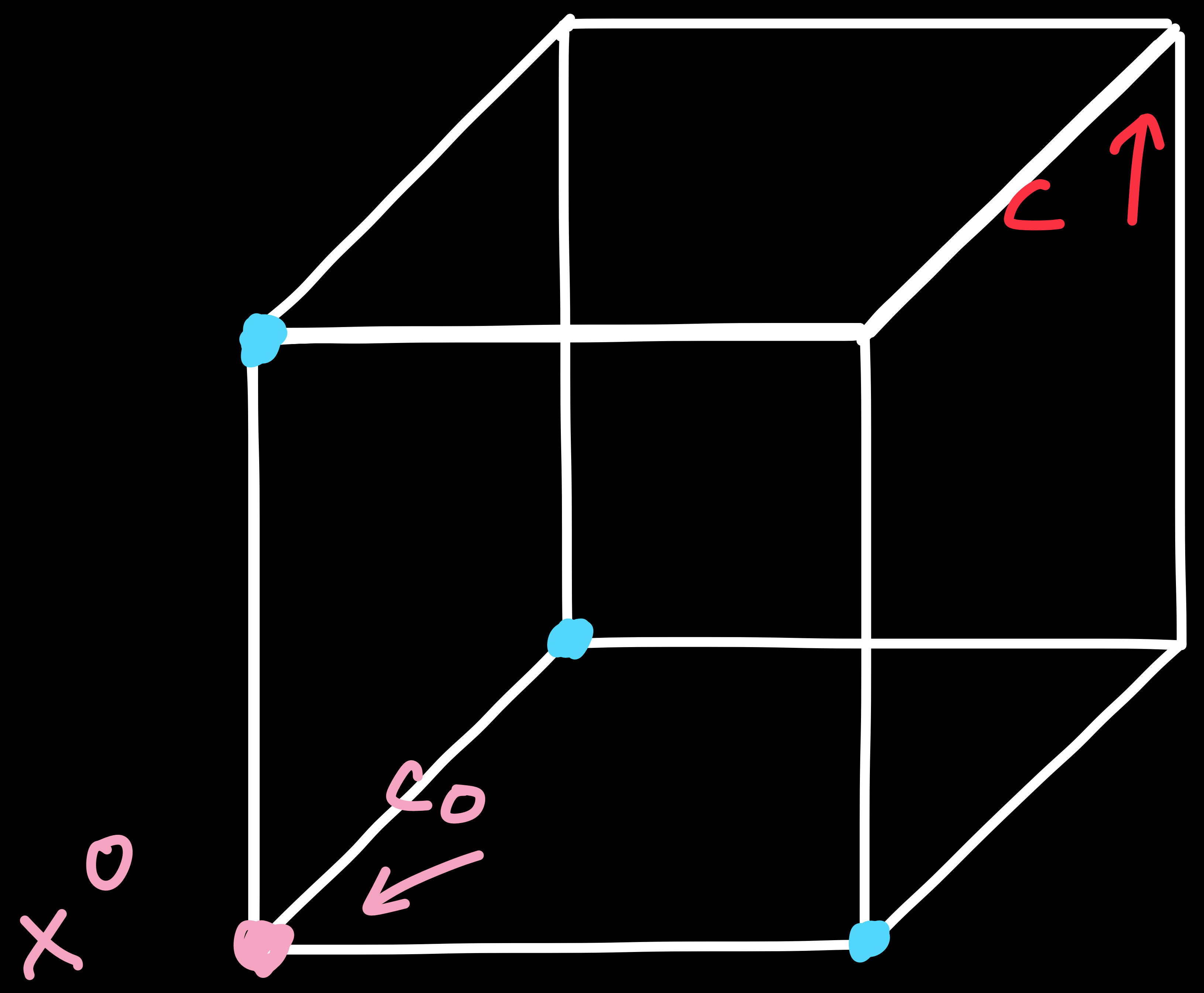


Simplex method



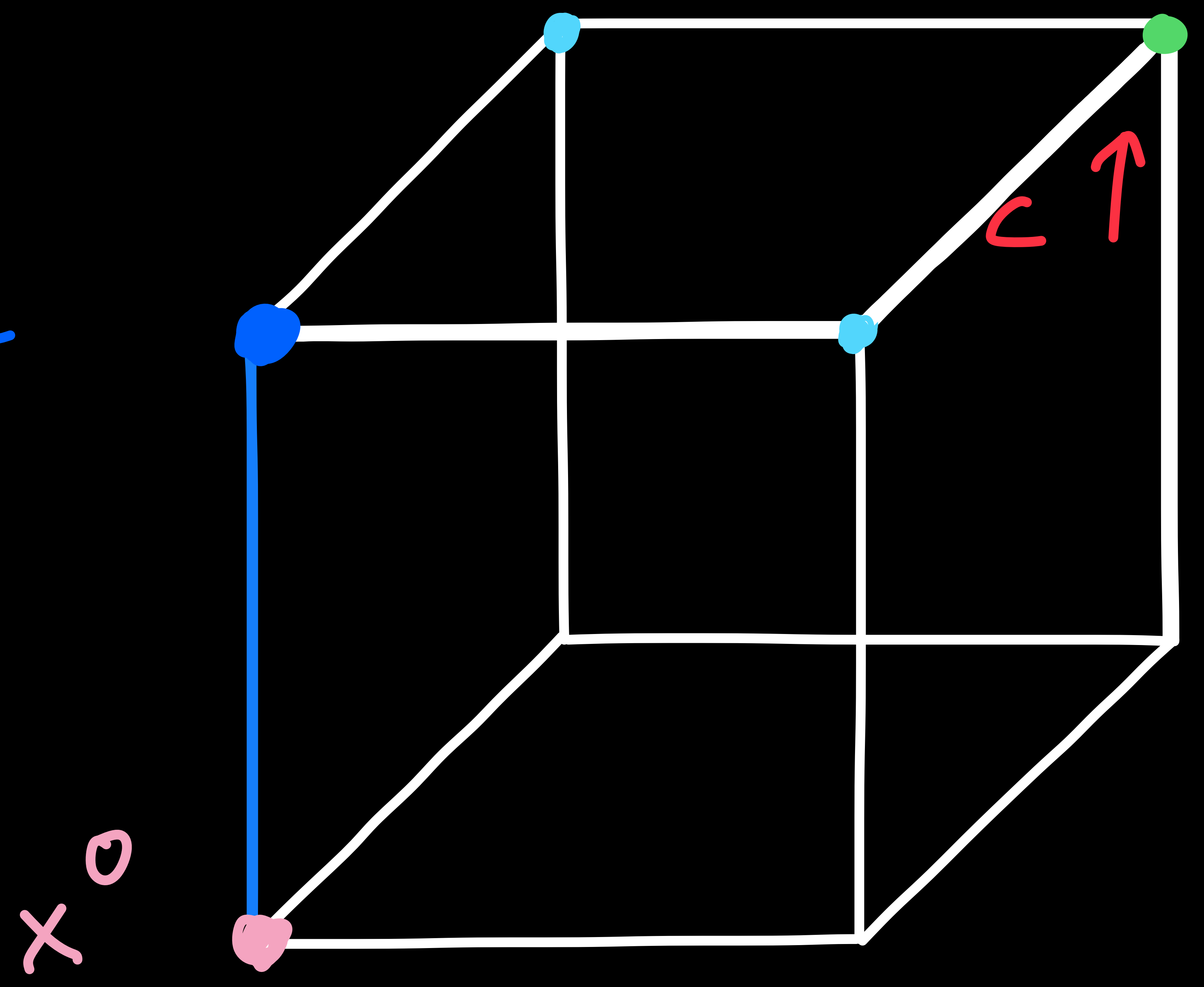
Phase 1

Simplex method

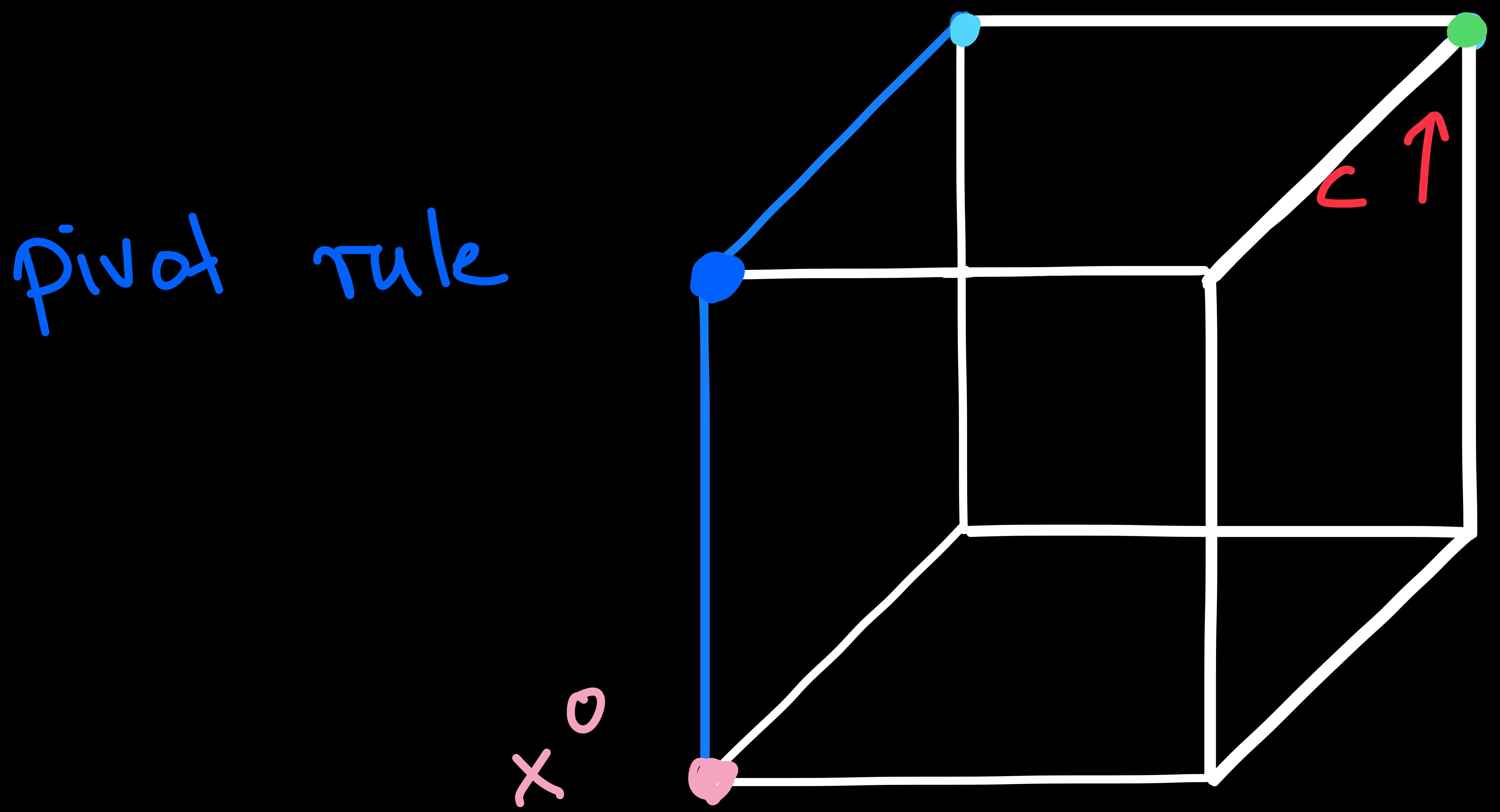


Simplex method

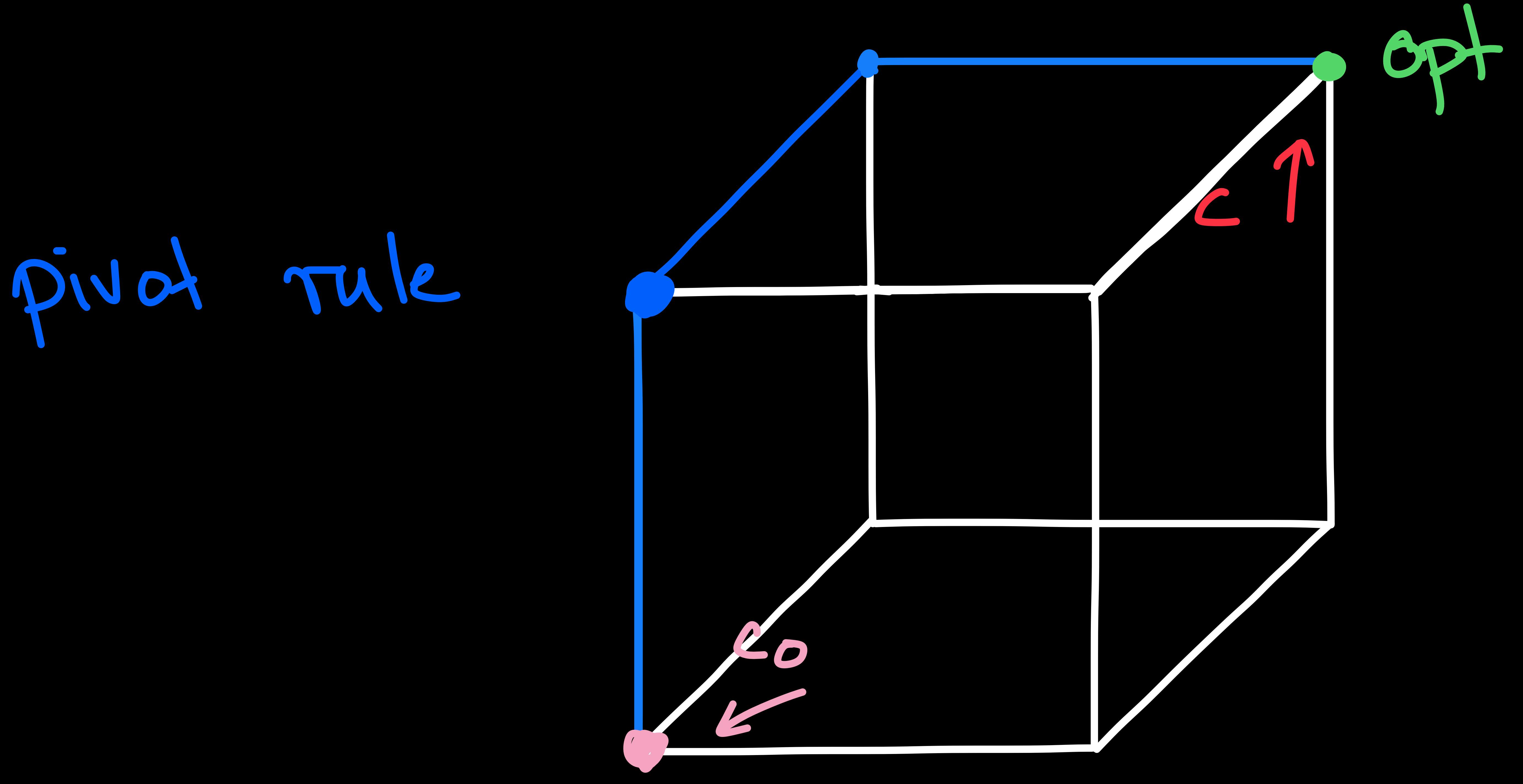
pivot rule



Simplex method



Simplex method



Runtime

The *Klee-Minty cubes* [22] [29] are the polytopes of the linear programs in $\mathbb{R}^n$ with $m = 2n$ facets given by

$$\begin{aligned} \min\ x_n : & \\ 0 \leq x_1 &\leq 1, \\ \varepsilon x_{i-1} \leq x_i &\leq 1 - \varepsilon x_{i-1} \end{aligned}$$

for $2 \leq i \leq n$ and $0 < \varepsilon < 1/2$. Our illustration shows the 3-dimensional Klee-Minty cube for $\varepsilon = 1/3$.

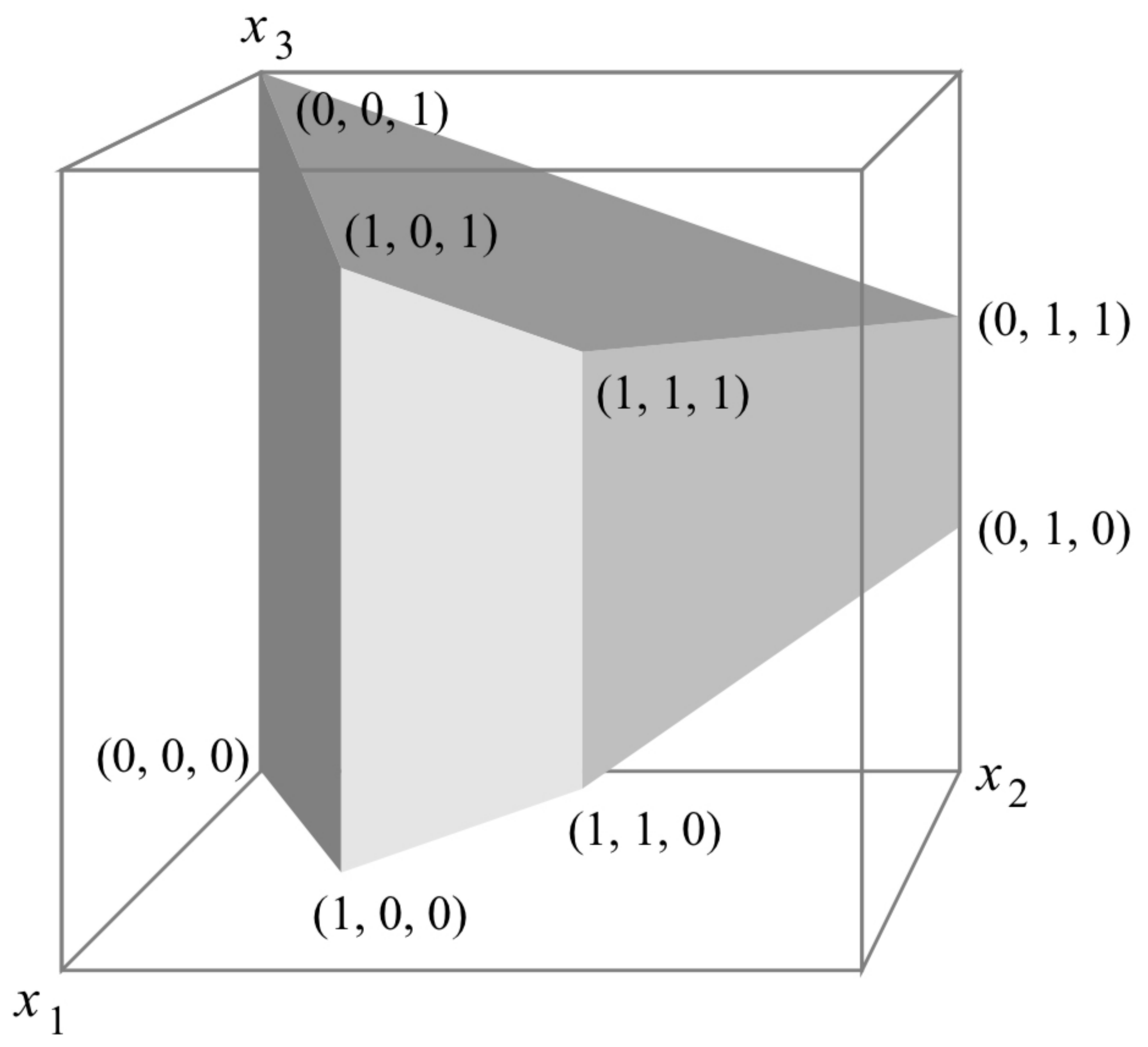


Figure 1: Klee-Minty cube for $n = 3, \varepsilon = 1/3$

Considering the geometry in the limit $\varepsilon \to 0$, one sees that the feasible region is a (slightly) deformed unit cube.

from : 'Randomized Simplex Algorithms on Klee-Minty-Cubes' by B. Gärtner, M. Henk, G.M. Ziegler

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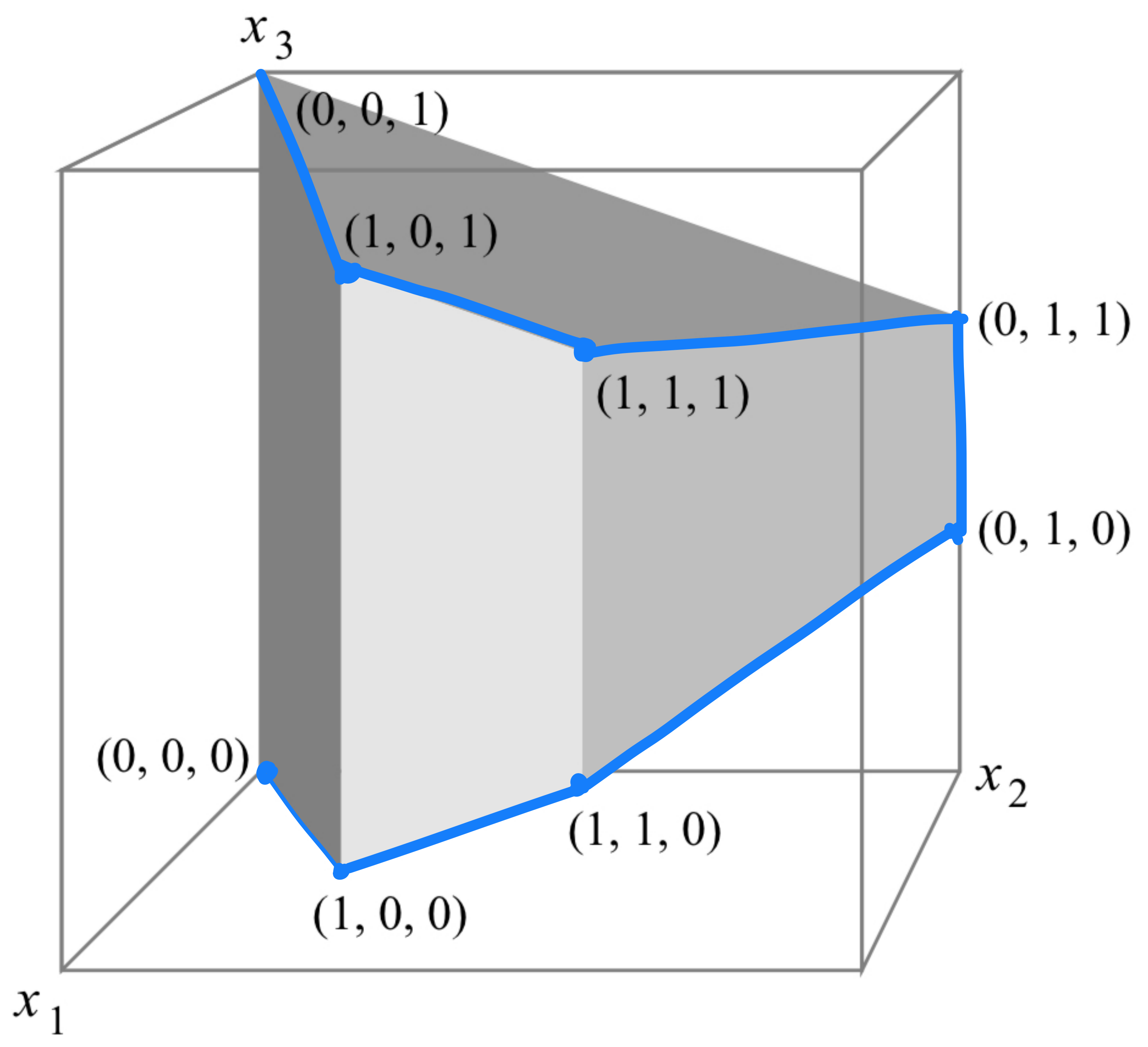

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Can we always "fool" simplex

like this ?

- for linear objectives c unknown

- for linear objectives c unknown
- for quadratic objectives answer is yes

- for linear objectives c unknown
- for quadratic objectives answer is yes:

"An unconditional lower bound for the active-set method in convex quadratic maximization"

joint work with Yann Disser, Sophie Huiberts and Nils Mosis

Algorithm 1: $\textsc{Simplex}(\boldsymbol{c}, A, \boldsymbol{b}, \boldsymbol{x})$

input: $\boldsymbol{c} \in \mathbb{Q}^n$, vertex $\boldsymbol{x}$ of polytope $\mathcal{P} := \{\boldsymbol{y} \colon A\boldsymbol{y} \leq \boldsymbol{b}\}$,
  system $A\boldsymbol{y} \leq \boldsymbol{b}$ non-degenerate
output: optimum solution $\boldsymbol{x}$ of $\max\{\boldsymbol{c}^\top \boldsymbol{y} \colon \boldsymbol{y} \in \mathcal{P}\}$

$\mathcal{B} \leftarrow \mathrm{eq}(\boldsymbol{x})$ `(basis)`
while $\mathcal{D} := \{\boldsymbol{d} \colon A_{\mathrm{eq}(\boldsymbol{x})}.\boldsymbol{d} \leq \boldsymbol{0},\ \boldsymbol{c}^\top \boldsymbol{d} > 0\} \neq \emptyset$**:**
  take $\boldsymbol{d} \in \mathcal{D}$ with $|\{i \in \mathcal{B} \colon \boldsymbol{A_i.d} = 0\}| = n-1$
  if $A_{\mathcal{B}}.\boldsymbol{d} \neq \boldsymbol{0}$**:** `(always true)`
    $\mathcal{B} \leftarrow \mathcal{B} \setminus \{\ell\}$ for the unique $\ell \in \mathcal{B}$ with $\boldsymbol{A_\ell.d} < 0$
  if $A_{\mathcal{B}}.\boldsymbol{d} = \boldsymbol{0}$**:** `(always true)`
    $\mu \leftarrow \inf\{\mu \geq 0 \colon \boldsymbol{x} + \mu\boldsymbol{d} \notin \mathcal{P}\}$
    $\boldsymbol{x} \leftarrow \boldsymbol{x} + \mu\boldsymbol{d}$
    if $\boldsymbol{c}^\top \boldsymbol{d} > 0$**:** `(always true)`
      $\mathcal{B} \leftarrow \mathcal{B} \cup \{j\}$ for the unique $j \in \mathrm{eq}(\boldsymbol{x}) \setminus \mathcal{B}$

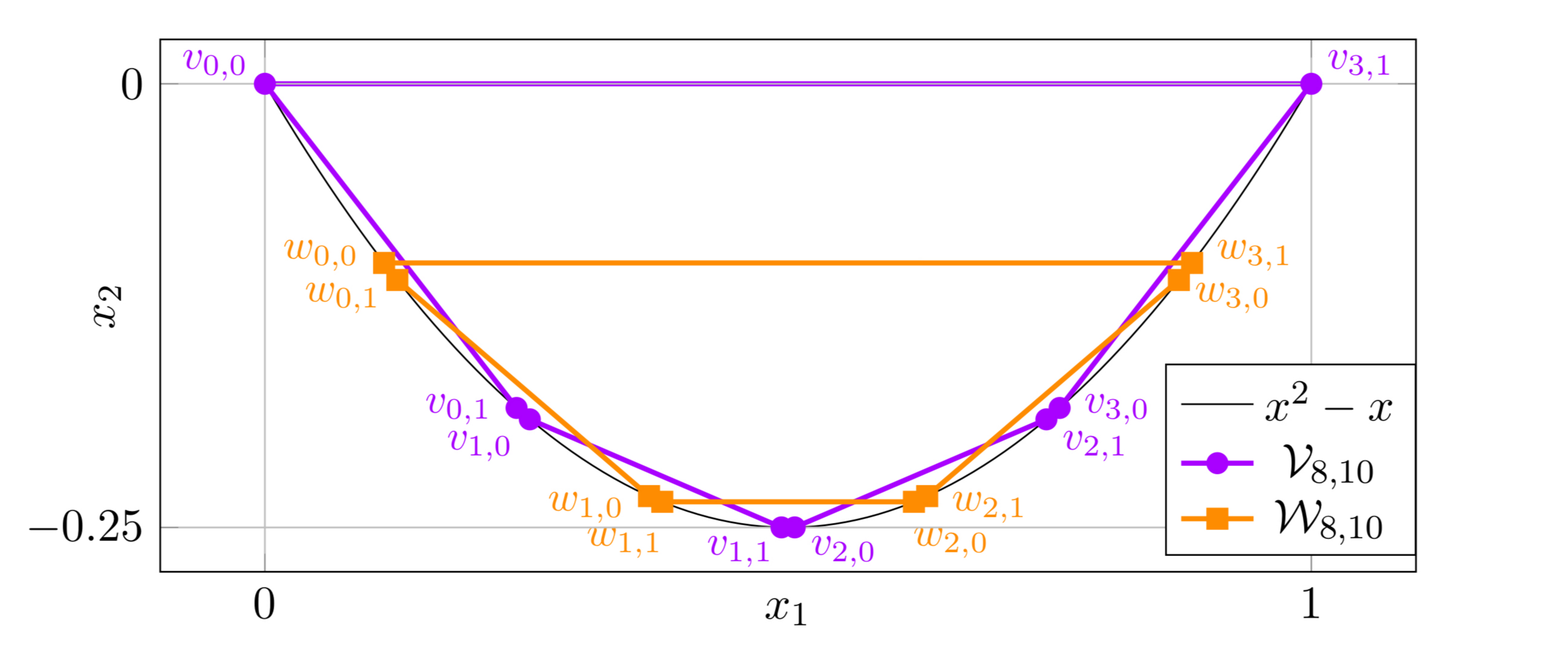
Algorithm 2: $\textsc{ActiveSet}(f, A, \boldsymbol{b}, \boldsymbol{x})$

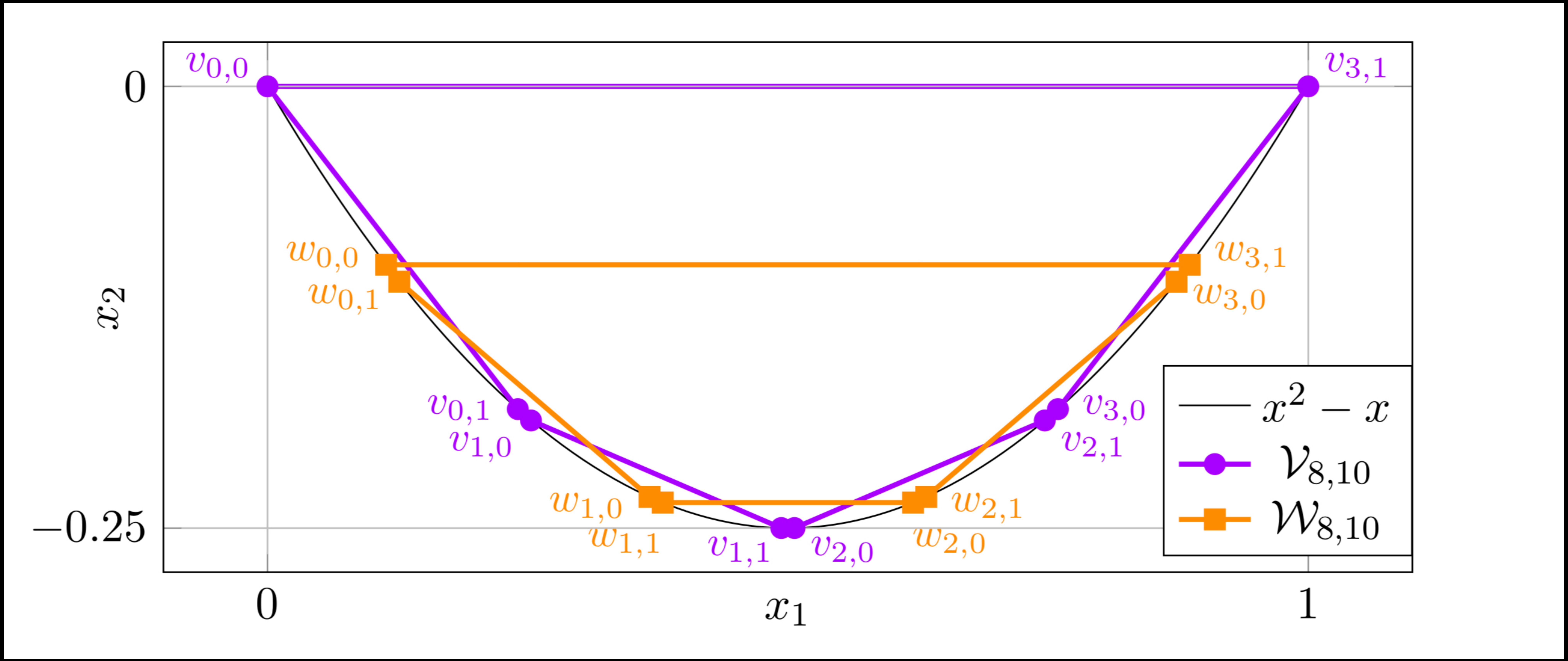
input: $f \colon \mathbb{R}^n \to \mathbb{R}$ continuously differentiable,
  point $\boldsymbol{x}$ of polytope $\mathcal{P} := \{\boldsymbol{y} \colon A\boldsymbol{y} \leq \boldsymbol{b}\}$
output: critical point $\boldsymbol{x}$ of $\max\{f(\boldsymbol{y}) \colon \boldsymbol{y} \in \mathcal{P}\}$

$\mathcal{A} \leftarrow \mathrm{eq}(\boldsymbol{x})$ `(active set)`
while $\mathcal{D} := \{\boldsymbol{d} \colon A_{\mathrm{eq}(\boldsymbol{x})}.\boldsymbol{d} \leq \boldsymbol{0},\ \boldsymbol{\nabla f(x)}^\top \boldsymbol{d} > 0\} \neq \emptyset$**:**
  take $\boldsymbol{d} \in \mathcal{D}$ maximizing $|\{i \in \mathcal{A} \colon \boldsymbol{A_i.d} = 0\}|$
  if $A_{\mathcal{A}}.\boldsymbol{d} \neq \boldsymbol{0}$**:**
    $\mathcal{A} \leftarrow \mathcal{A} \setminus \{i\}$ for some $i \in \mathcal{A}$ with $\boldsymbol{A_i.d} < 0$
  if $A_{\mathcal{A}}.\boldsymbol{d} = \boldsymbol{0}$**:**
    $\mu \leftarrow \inf\{\mu \geq 0 \colon \boldsymbol{x} + \mu\boldsymbol{d} \notin \mathcal{P}$ or $\boldsymbol{\nabla f(x + \mu d)}^\top \boldsymbol{d} \leq 0\}$
    $\boldsymbol{x} \leftarrow \boldsymbol{x} + \mu\boldsymbol{d}$
    if $\boldsymbol{\nabla f(x)}^\top \boldsymbol{d} > 0$**:**
      $\mathcal{A} \leftarrow \mathcal{A} \cup \{j\}$ for some $j \in \mathrm{eq}(\boldsymbol{x}) \setminus \mathcal{A}$

Disser & Mosis '25

$$f((x_1, x_2)) = x_1^2 - c\,x_1 - x_2$$





- extended formulation via deformed products

(Amenta & Ziegler '99)

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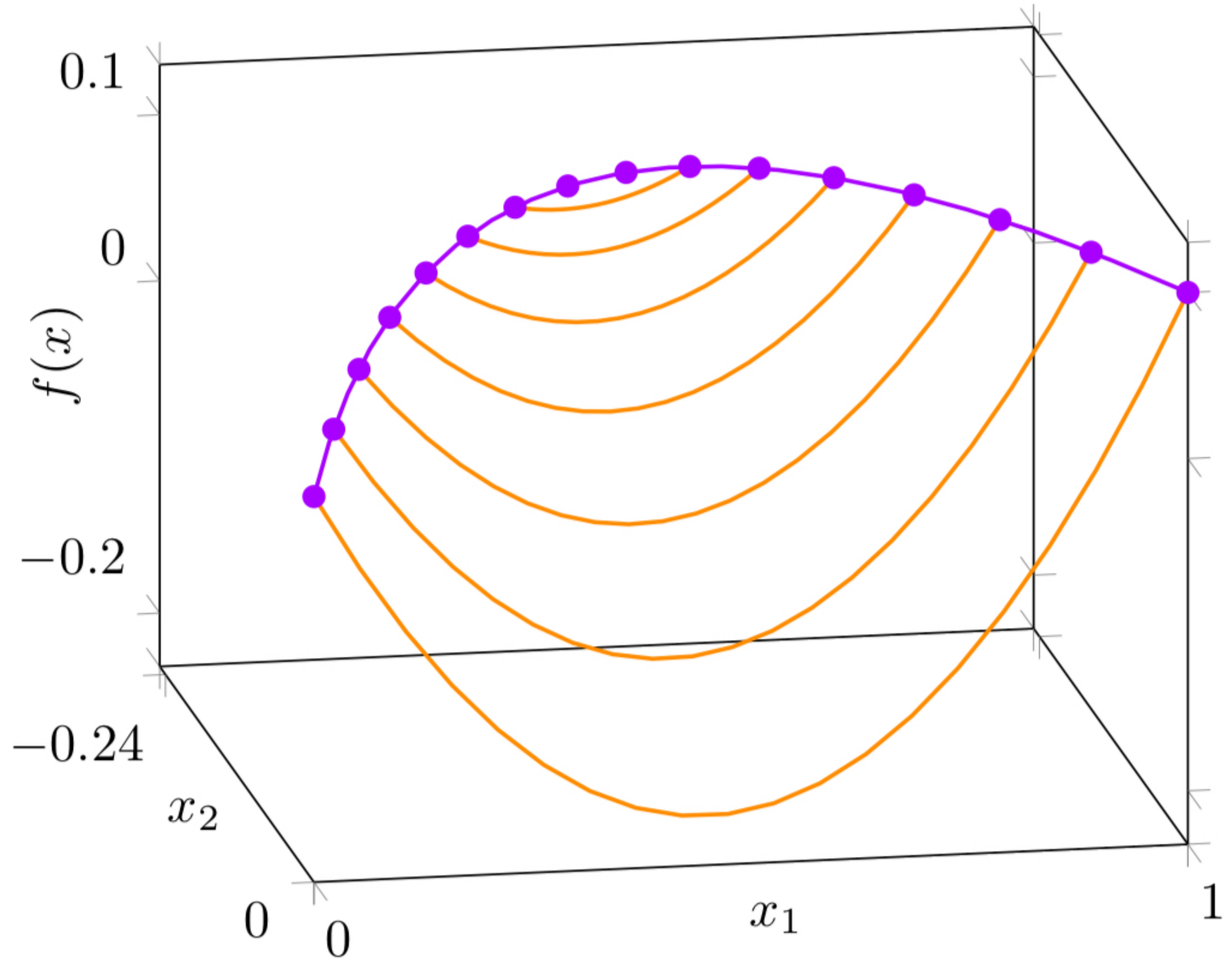


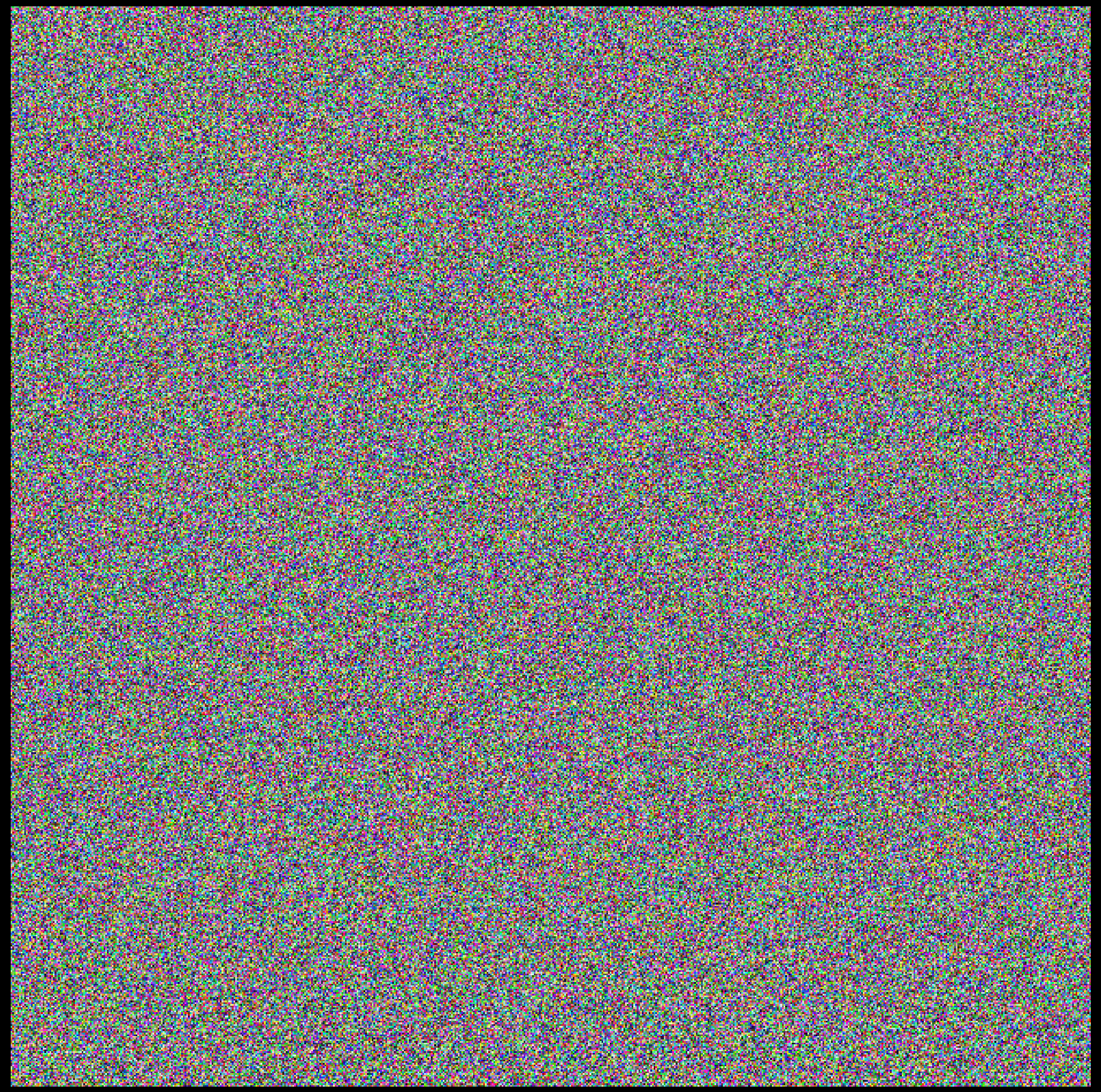
Figure 4: Objective function values along vertices and edges of the projection $\Pi(\mathcal{P})$ in (QP) for $n = M = 16$, $d = 4$. Purple edges are monotonically increasing in objective value, while the orange edge and chords have negative gradients at both endpoints.

Practice: run time linear in n and d.

How to explain / model this?

Worst case:

average case:



Worst case:

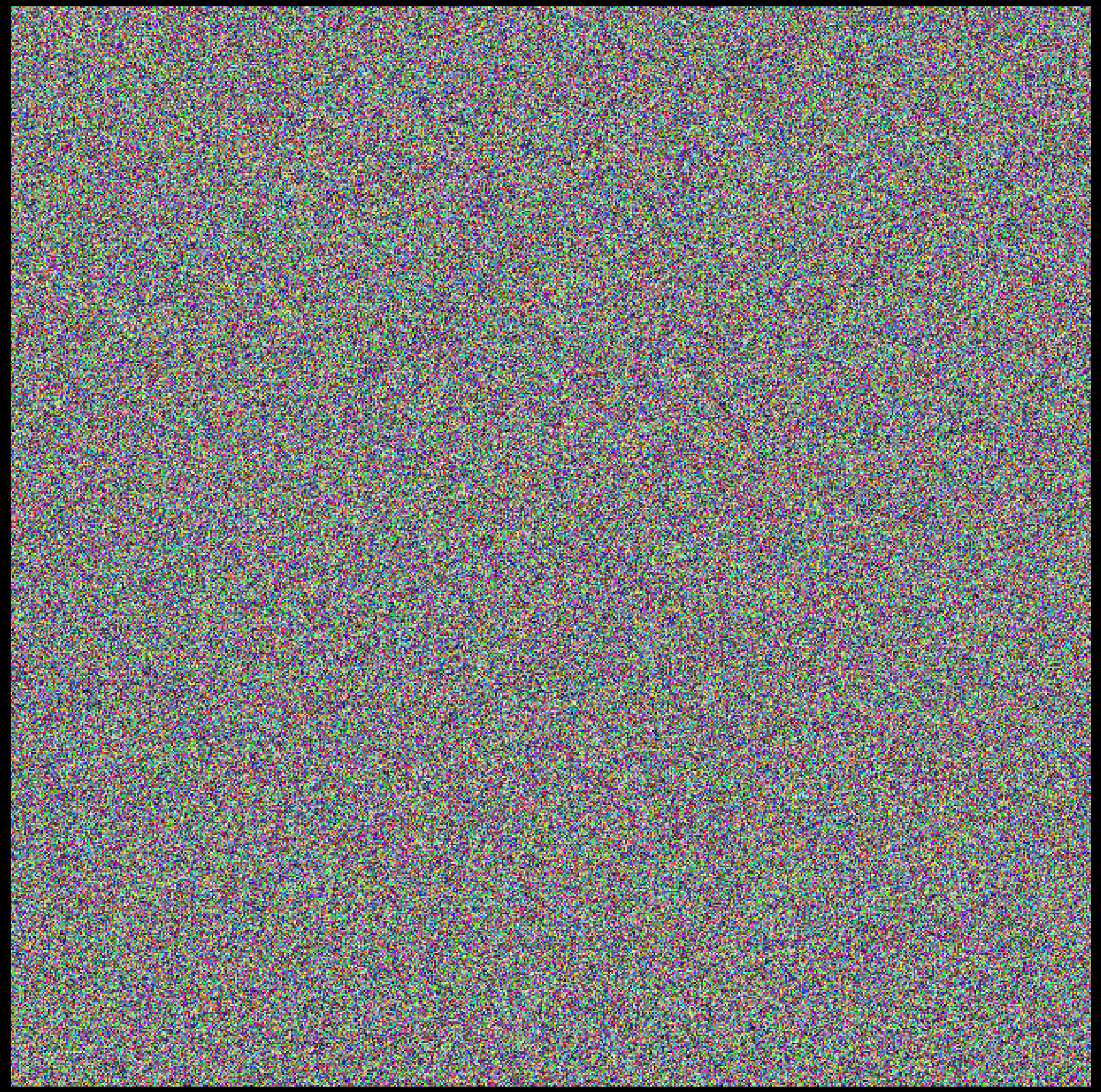
structure ✓

upper bounds
(poly) ✗

average case:

structure ✗

upper bounds
(poly) ✓



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0 σ ∞

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Smoothed analysis.

structure ✓

upper bounds (poly) ✓

average case:

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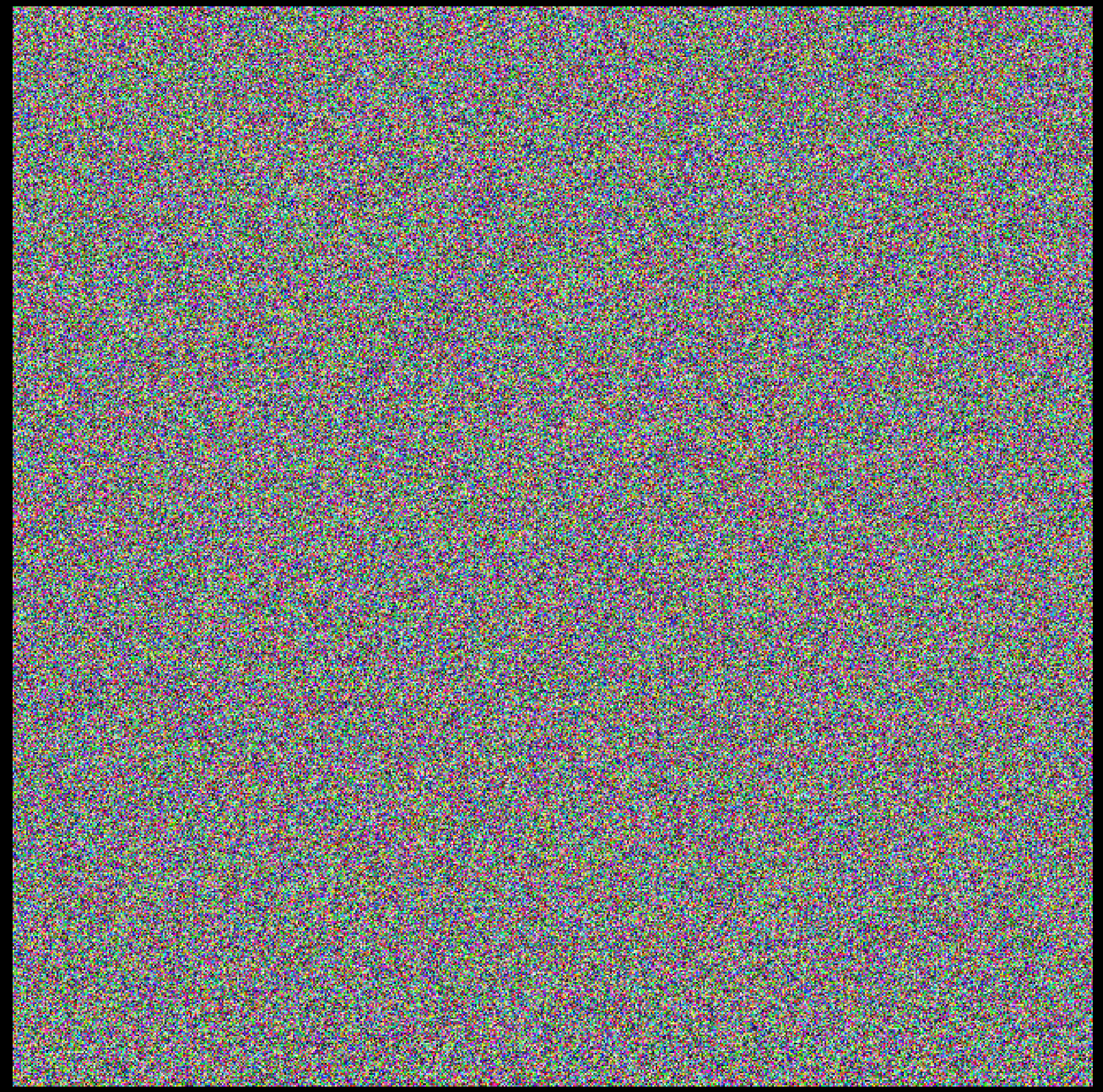
upper bounds (poly) ✓



0 σ ∞

Smoothed analysis

Let $\bar{A} \in \mathbb{R}^{n \times d}$ have rows of norm ≤ 1.

$\bar{b} \in [-1, 1]^n$, $c \in \mathbb{R}^d$

Let $\hat{A}, \hat{b}$ have i.i.d. $N(0, \sigma^2)$ entries

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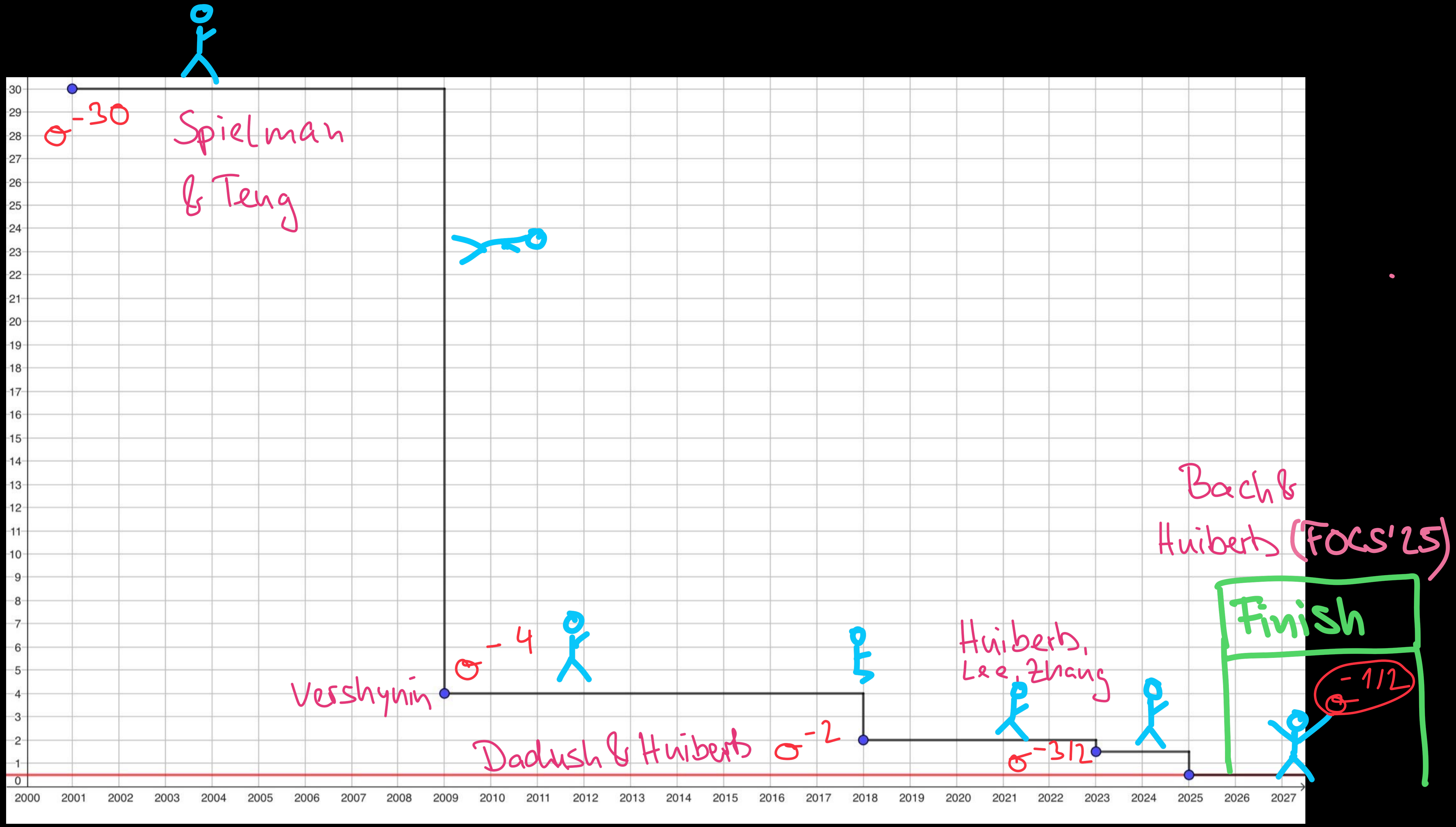
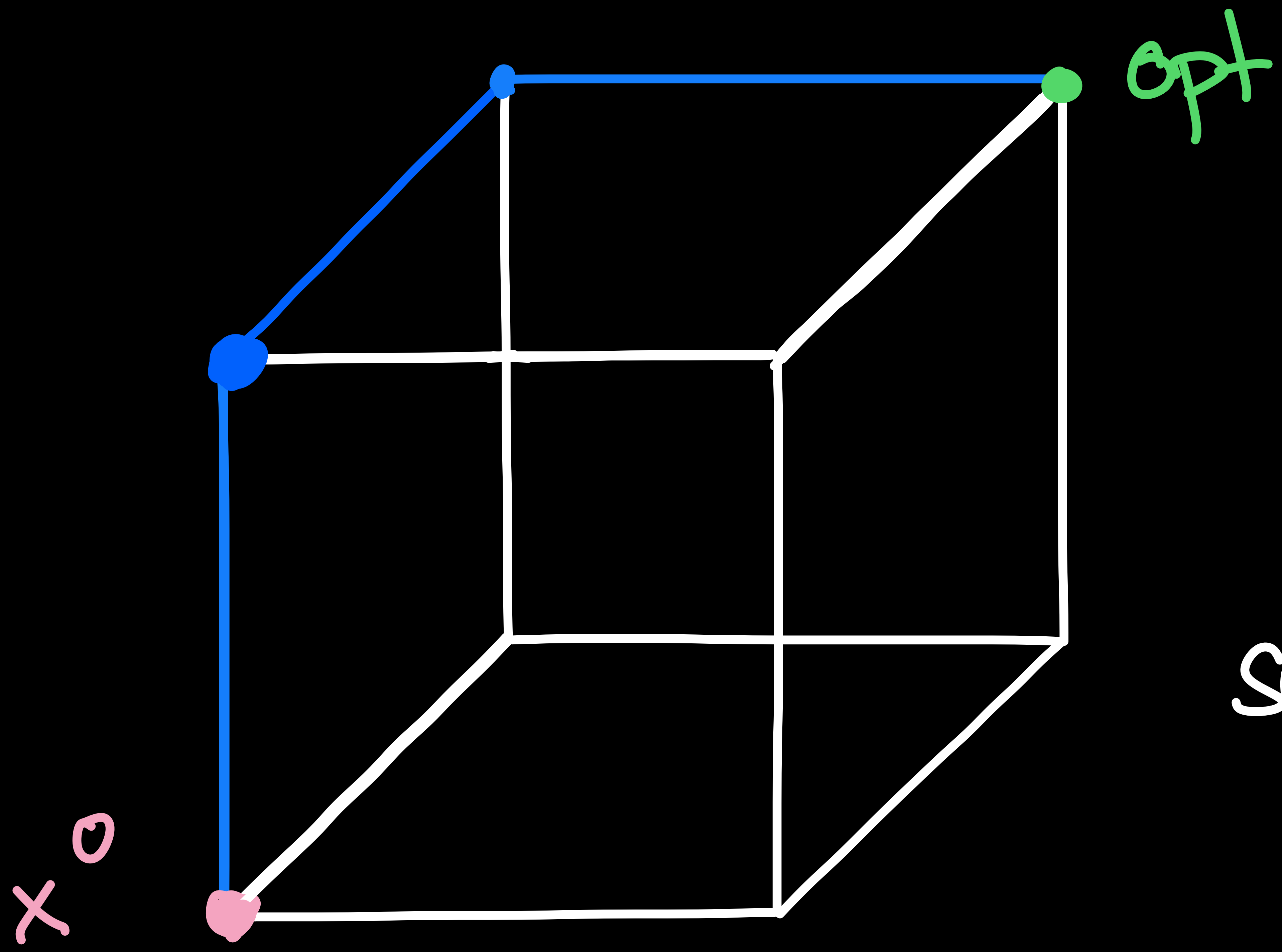
Theorem:

$$\max_{\bar{A}, \bar{b}, c} \mathbb{E}_{\hat{A}, \hat{b}} \left[\begin{array}{l} \text{time to solve} \\ \quad \max \; x \\ \text{s.t. } (\bar{A} + \hat{A}) x \leq \bar{b} + \hat{b} \end{array} \right] = \sqrt{\sigma^{-1}} \cdot \mathrm{poly}(d, \log n)$$



σ^-30
Spielman & Teng
Vershynin
σ^-4
Dadush & Huiberts σ^-2
Huiberts, Lee, Zhang
σ^-3/2
Bach & Huiberts (FOCS'25)
Finish
σ^-1/2

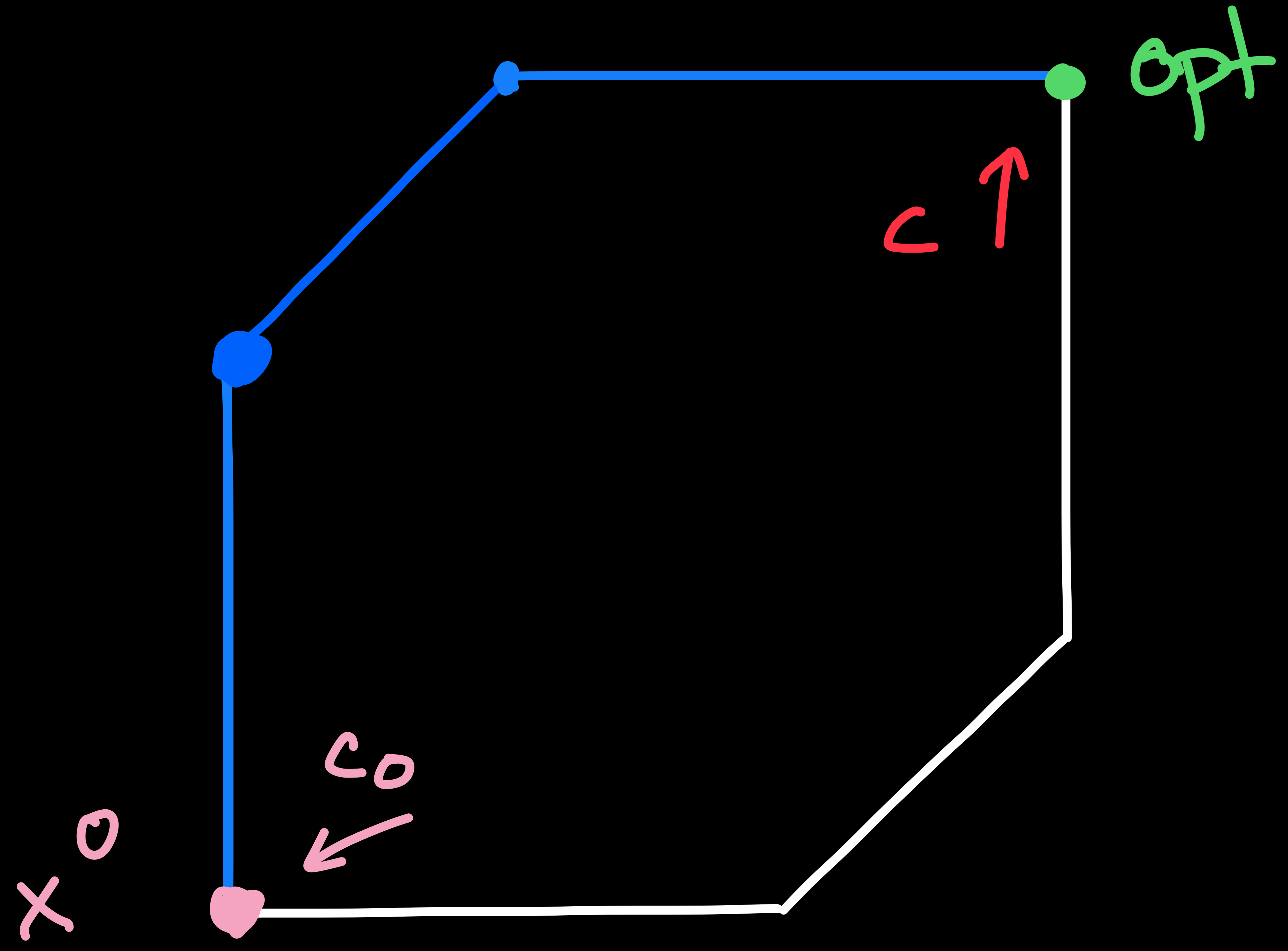
Shadow vertex simplex method



project to

$\longrightarrow$

$\mathrm{span}(c_0, c)$



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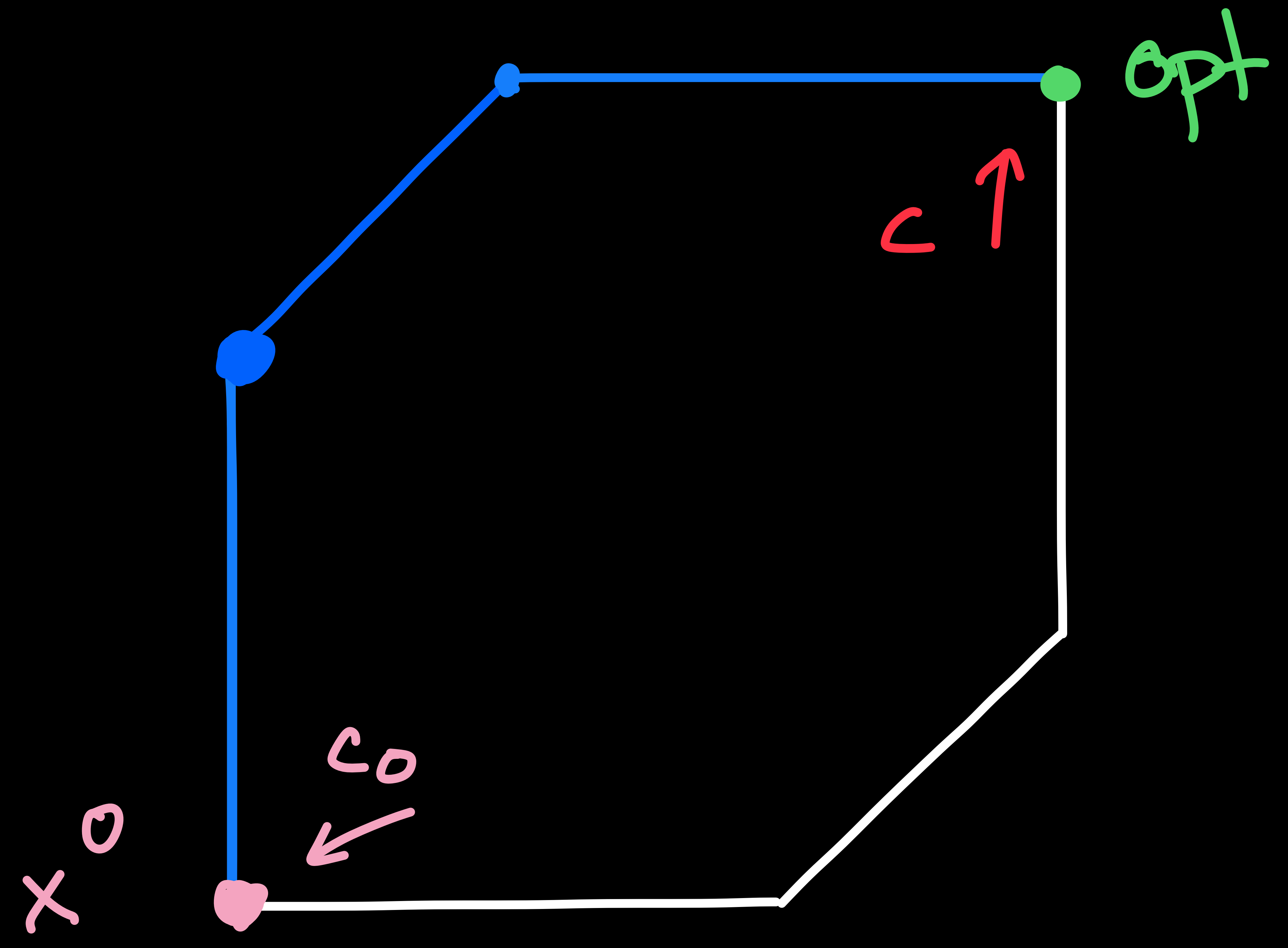

- solutions to
  $$\max\ (\lambda c_0 + (1-\lambda)c)^T x$$
  $$\text{s.t. } Ax \leq b$$
  form simplex path

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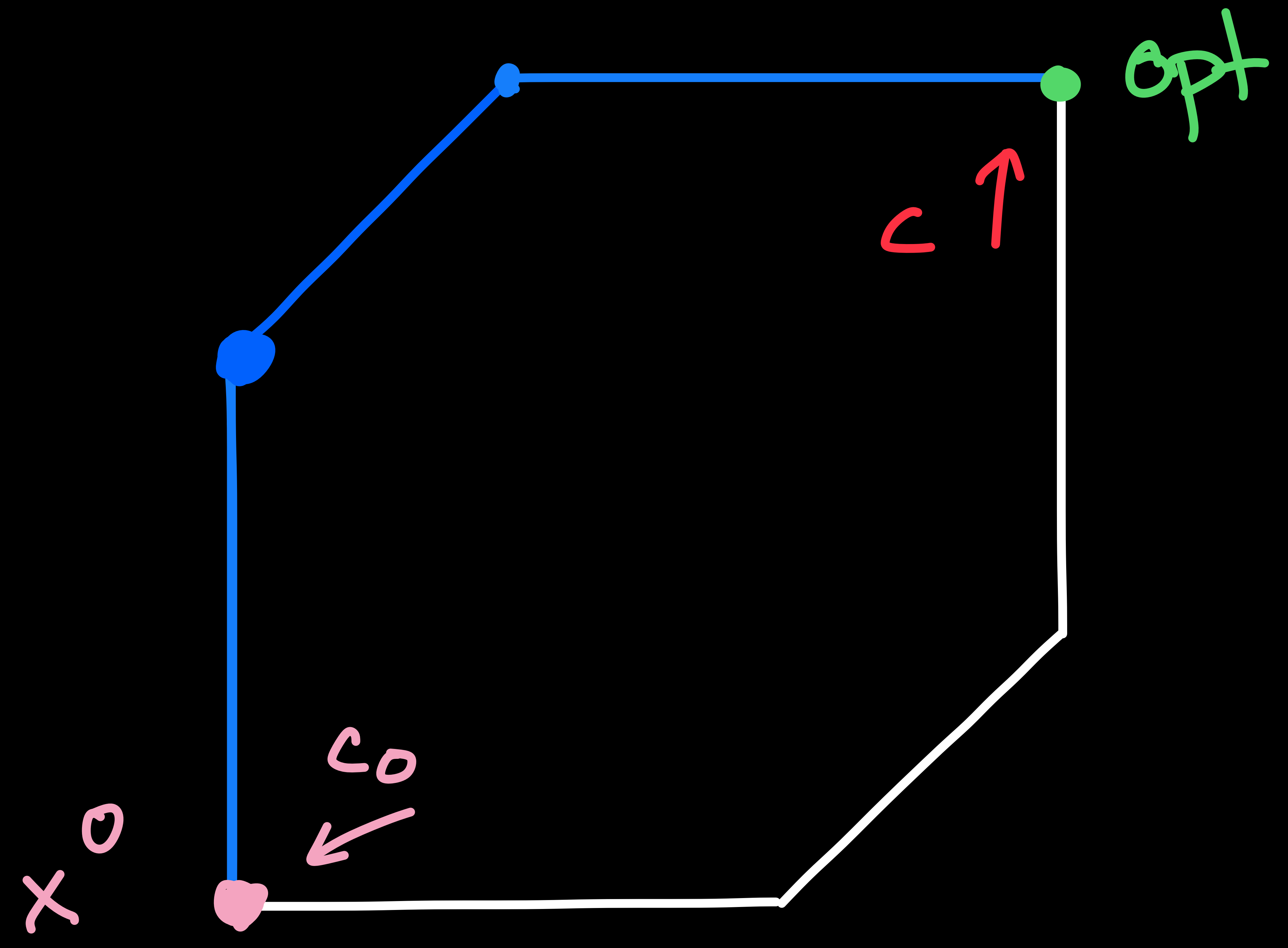

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form simplex path

→ we bound # vertices on path

How to bound # vertices ?

1. Choose quantities to measure on $\mathbb{R}^d$-polytope
2. project $\{x : Ax \le b\}$ to span(c_0, c)
3. choose potentials

   perimeter | exterior angle
   polar perimeter | polar exterior angle
   etc.

4. potential measure

Our improvements

1. slack $(Ax \le b \to Ax + s = b)$, multipliers (normal cone

$$N(x) = \{ y \in \mathbb{R}^d : y = \sum \lambda_i a_i, \lambda_i \ge 0 \} = \{ y \in \mathbb{R}^d : A_x^{-1} y \ge 0 \})$$

$\ge m$

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2. pick c_0 random, c fixed (adapt algo reduction)

$\downarrow$

independent of $\bar{A}, \bar{b}$ (and c)

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3. pick two clever potentials
"perimeter integral" & "exterior angle"
4. combine them to vertex-neighbor separation
→ each step consums a lot of either integral

Does smoothed analysis explain efficiency of the simplex method?

Problem 1:

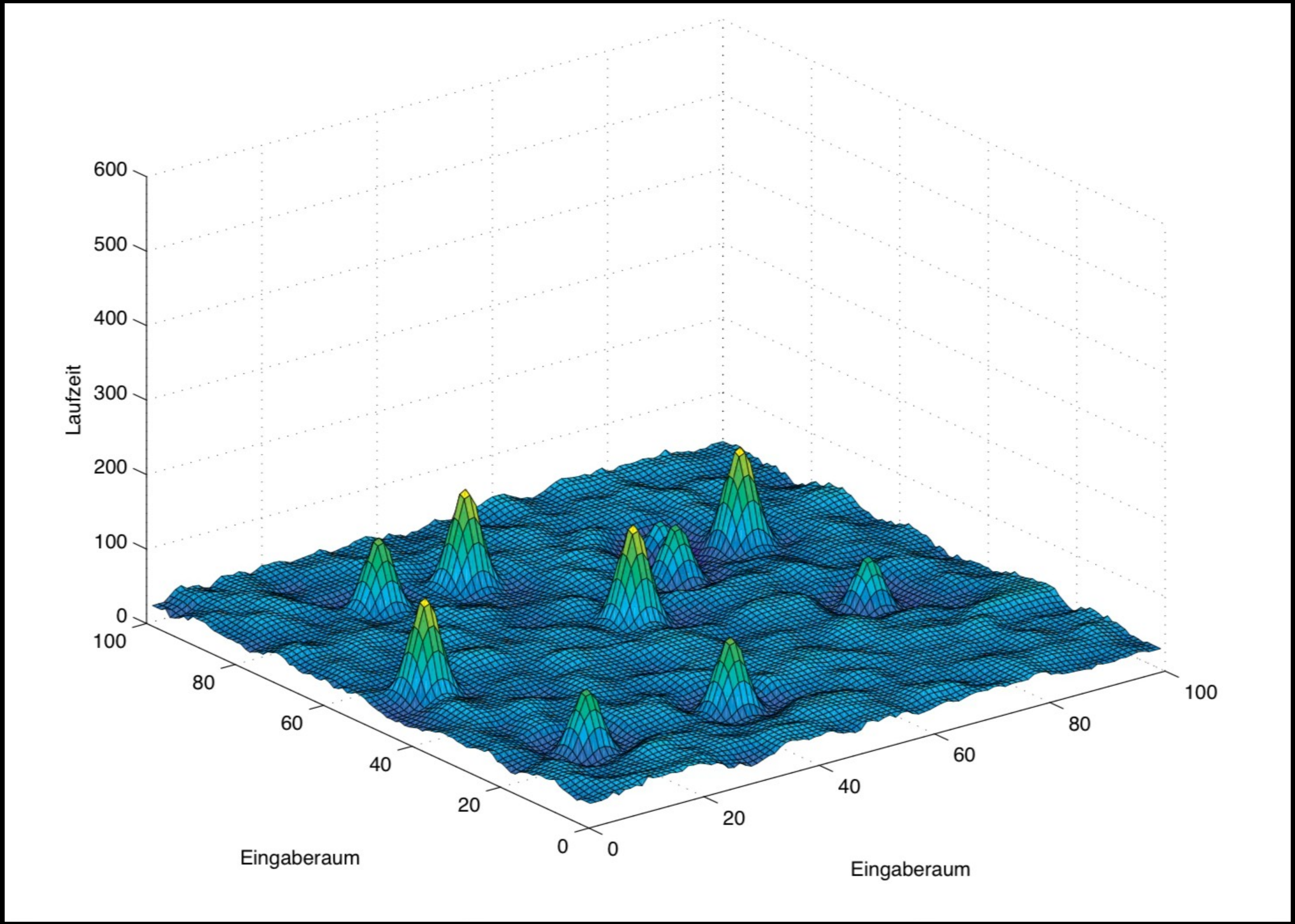
Real world LP:

$< 0.1\%$ of $A_{i,j} \neq 0$

Smoothed analysis

all $A_{i,j} \neq 0$

Problem 2:



K. H. Borgwardt: "Wie schnell arbeitet das Simplexverfahren normalerweise?"

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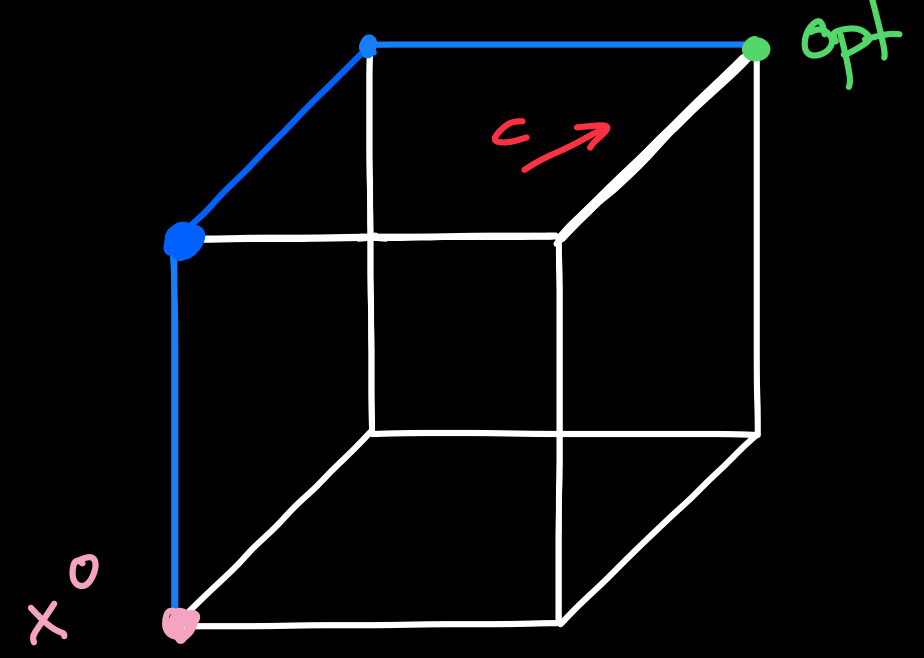
What are we modelling? What is σ?

- measurement error? $\to 10^{-2}$ sigfigs accuracy
- brittleness? $\to \frac{o(1)}{\sqrt{d}}$ more signal than noise?
- process within solver? $\to 10^{-6}$ solver tolerance?
- floating point imprecision? 10^{-16} 64-bit IEEE 754?

Are we studying the simplex method?

- This geometry  does not exist

- What exists: $\left\{ \text{numerical linear algebra} \right.$

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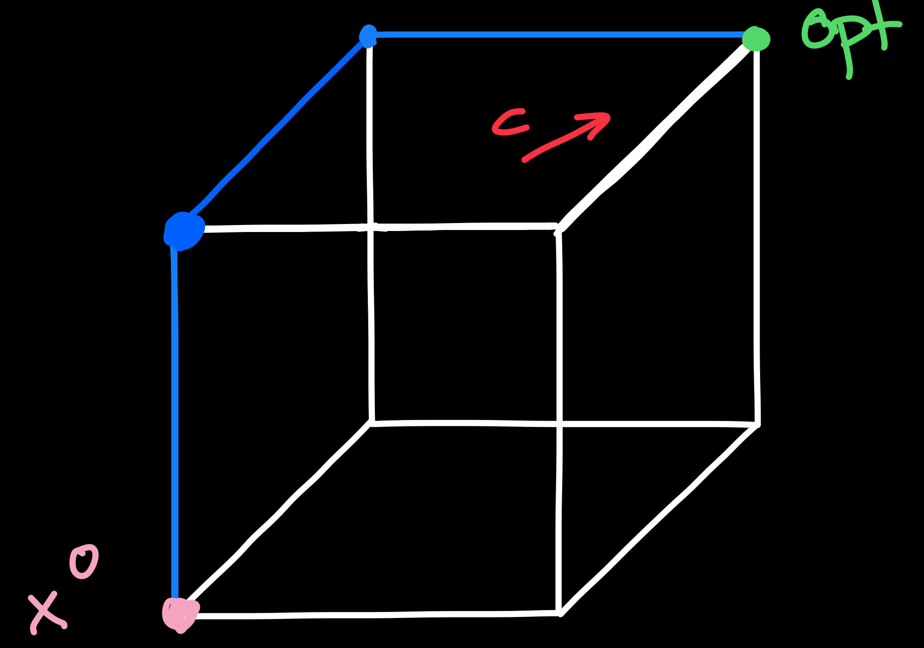
- This geometry  does not exist

- What exists: $\left\{ \begin{array}{l} \text{numerical linear algebra} \\ \text{bound shifting} \end{array} \right.$

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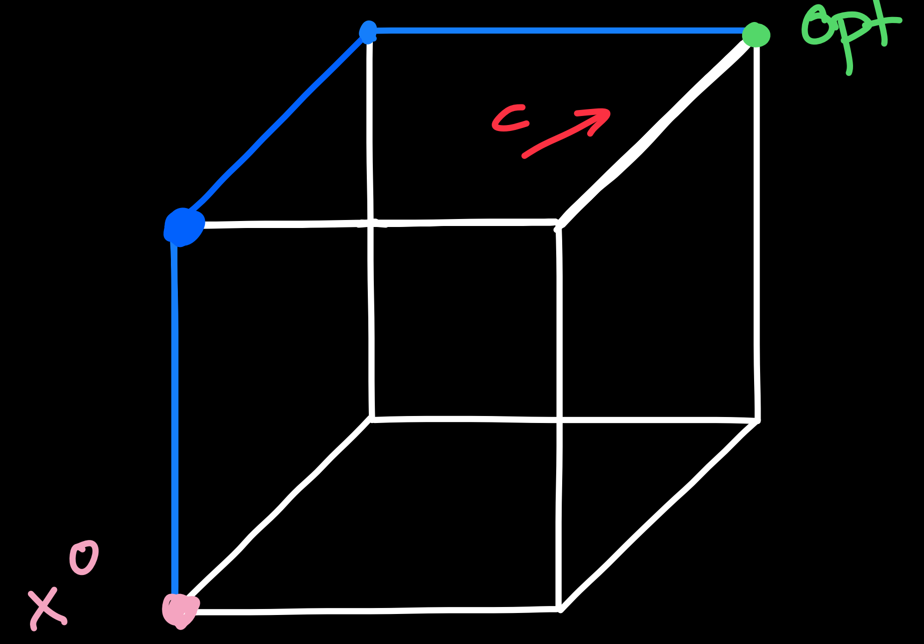
- This geometry 

does not exist

- What exists:
  - numerical linear algebra
  - bound shifting
  - bound perturbations

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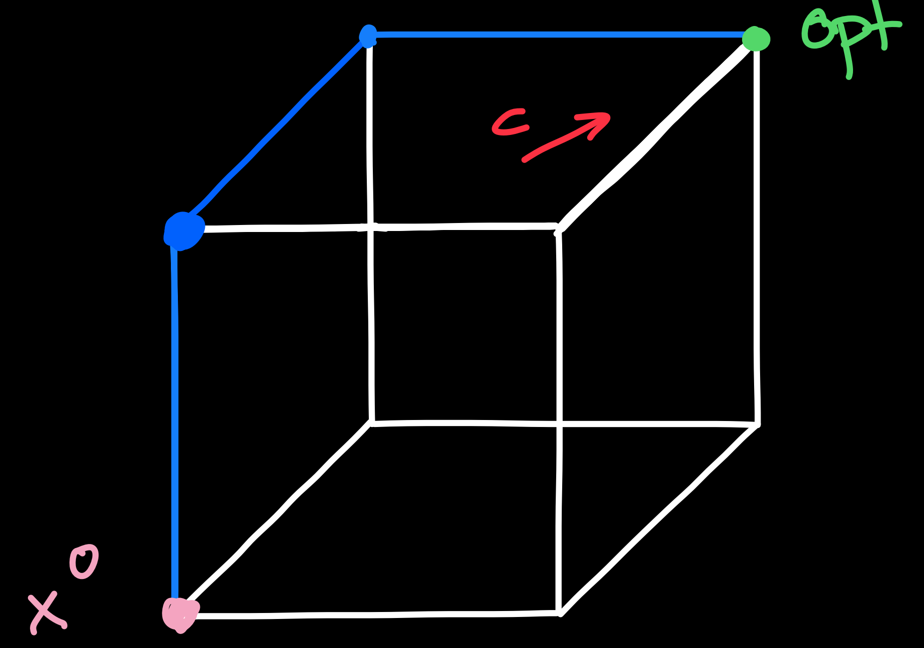
- This geometry  does not exist

- What exists: $\left\{\begin{array}{l}\text{numerical linear algebra}\\ \text{bound shifting}\\ \text{bound perturbations}\\ \text{(Harris) ratio test}\end{array}\right.$